

What is a signal?

— Signal is basically data.

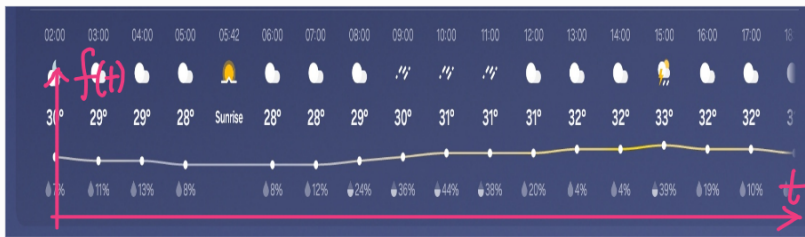
In mathematical terms any function of a variable can be a signal.

$f(x)$, $f(t)$ are signals that depend on one variable ← univariate

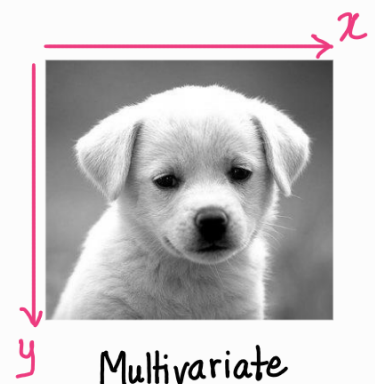
$f(x, y)$, $f(x, t)$ are signals that depend on multiple variables ← multivariate

example of univariate signal — temperature readings against time

example of multivariate signal — grayscale image



Univariate



Multivariate

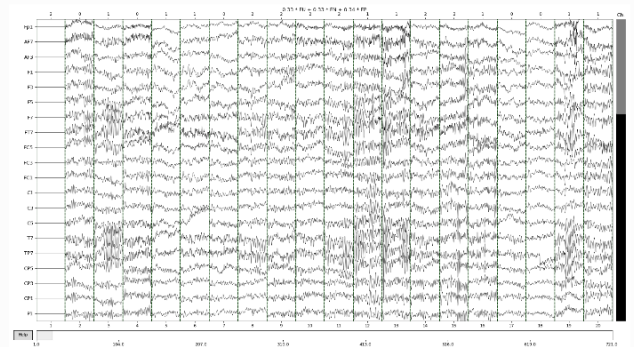
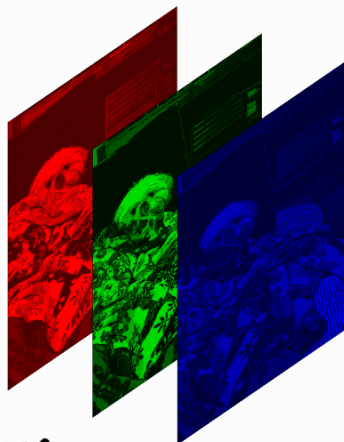
signals can be multi-channelled — aggregate of functions depending on same variables acting as complete info

for example — RGB images are 3-channelled signals

— typical EEG readings (>3 channels)



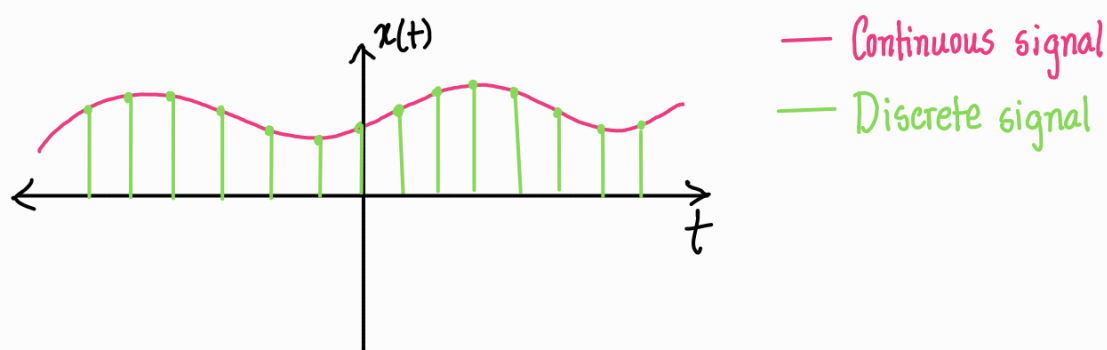
RGB image



EEG data

Signals in the wild vs Signals in our devices

At the origin all signals are continuous. By continuous we mean that $f(x)$ is defined for all values of $x \in \mathbb{R}$. For example if $f(x)$ is continuous, you will know the value of $f(x)$ at $x = 1.2345678$ units & also at $x = 1.23456789$ units. That is all fun until you have to store it in your finite space device. A continuous signal has infinite data points & it is just not possible to store such a signal in our physical devices. So instead of storing raw continuous data, we sample our signals to reduce the signal size.



One small note — From now on we will stick to time dependent signals for a while.

for time signals, continuous signals are called continuous time signals

discrete signals are called discrete time signals

The time difference between two samples is called 'sampling period'. We will refer to sampling period as T_s .

If the signal is uniformly sampled, time difference between all samples will be the same. We will assume uniform sampling throughout the course.

Representing discrete time signals —

notice how $x(t)$ is only defined

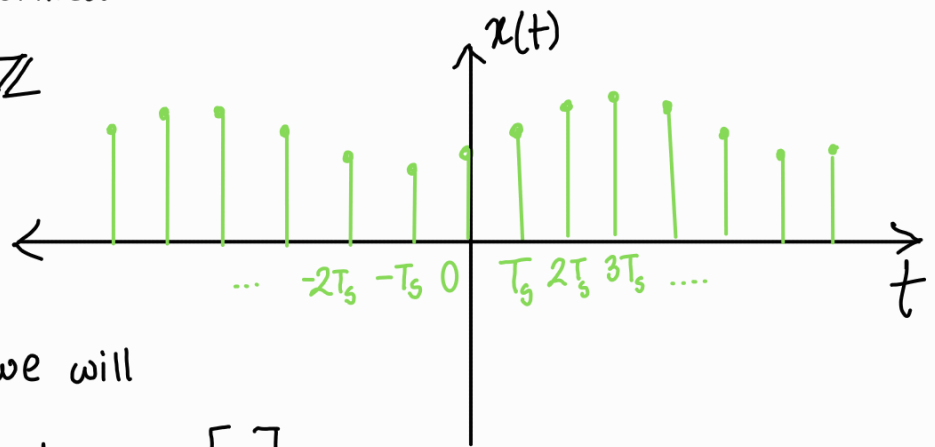
for $t \in n \cdot T_s$ where $n \in \mathbb{Z}$

So, to demarcate between

continuous time signals

& discrete time signals, we will

redefine discrete time signals as $x[n]$.

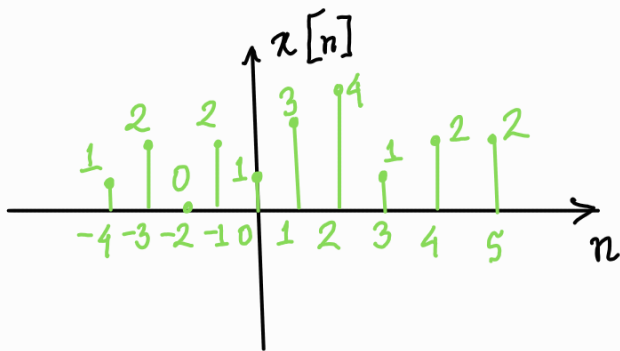


Then, $x[n] = x(nT_s)$. T_s is being considered implicitly for compactness.

discrete

continuous

We will also update the x-axis to n . For example, a discrete signal —



$x[n]$ can be represented as an array —

$$x[n] = \{1, 2, 0, 2, 1, 3, 4, 1, 2, 2\}$$

The arrow represents the value @ $n=0$.

It is standard to use curly braces.

Discrete signals vs Digital signals

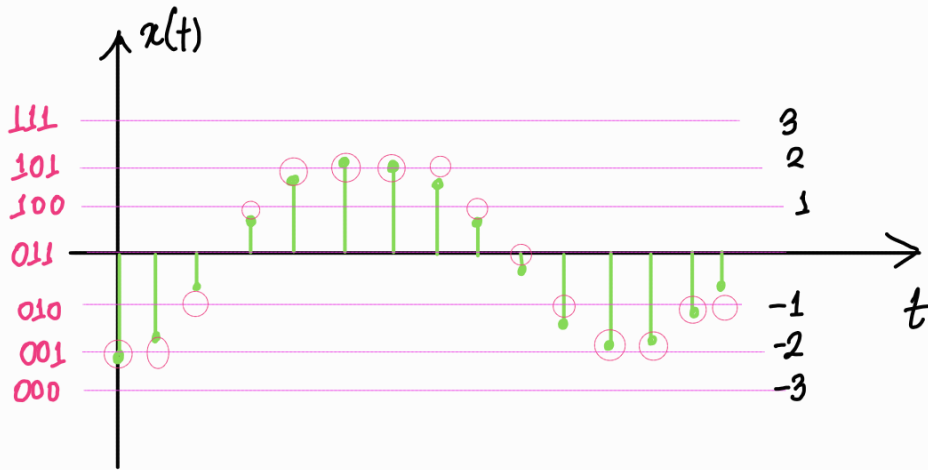
Discrete signals and digital signals are different. Digital signals are quantized, discrete signals are not. To point the difference quite simply —

* Discrete x/t axis — discrete signal

* Discrete x/t axis + discrete function values — digital signals

Quantization —

Since all digital signals are stored as binary, to store any signal digitally, we have to clamp the amplitude to discrete binary values. This can be considered as sampling in the amplitude axis. Let's look at an example —

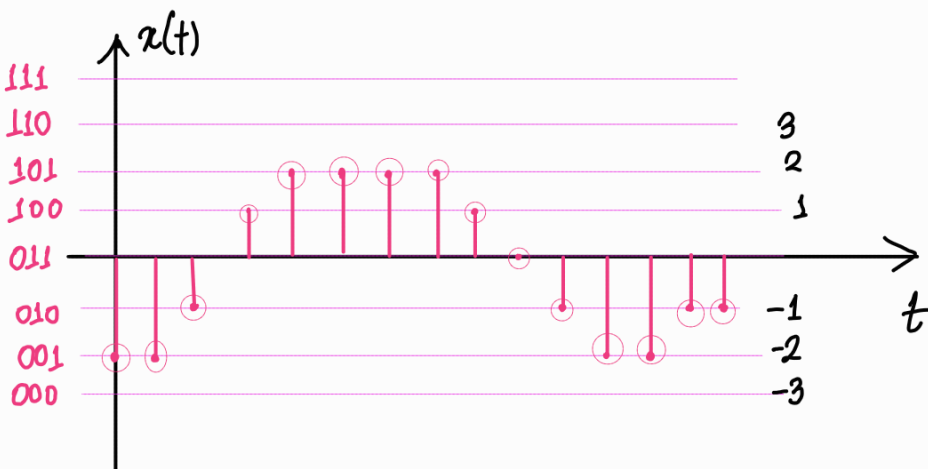


So clearly we have some discrepancies between the original signal level and the quantization levels. We call this quantization error.

But it is quite clear that if we increase our binary levels, we will for sure get less quantization error.

Here we have used 3 bit binary representation, which is very coarse.

For general purposes, we use higher bit representations. For example, CDs record audio 16 bits while high quality Blu-rays use 24 bits.



Some terminologies —

no. of bits used to quantize is called bit depth

no. of quantization levels = $2^{\text{bit depth}}$

difference between two quantization levels is called quantization step size and is given by —

$$\text{step size, } \Delta = \frac{\text{max level} - \text{min level}}{n(\text{levels}) - 1}$$

Do note that max quantized level may not be equal to $\max(x(t))$.

So in this course we will not go too deep into quantization and its effects. We will assume all our signals, when discrete, to be digital in nature. You will learn more about quantization and data encoding in CSE320, or you probably already have covered it. At best, I may give you a signal, with max and min quantization level with bit depth and ask you to quantize the signal.